

COSC 594/650

Convex optimization

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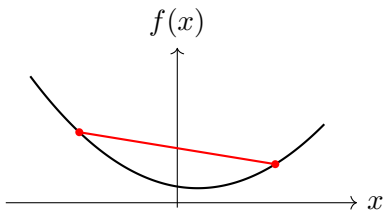
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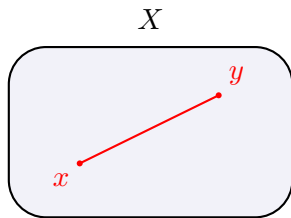
What makes an optimization problem convex?

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & x \in X\end{array}$$

Convex objective



Convex feasible set

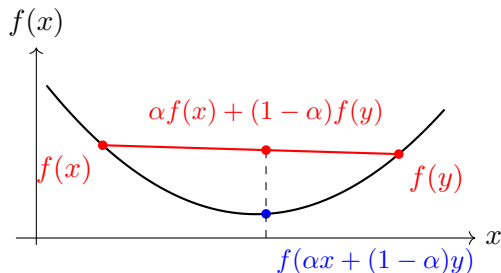


The Chord test

Convex function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \quad \forall x, y \in \mathbb{R}^n, \alpha \in [0, 1]$$

■ GEOMETRIC MEANING



★ epigraph $\{(x, t) \mid f(x) \leq t\}$ – convex set

Two practical convexity tests

Convex function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \quad \forall x, y \in \mathbb{R}^n, \alpha \in [0, 1]$$

■ CONVEXITY ALONG ALL LINES

$$\phi(t) = f(x_0 + th) \text{ is convex,} \quad \forall x_0, h \in \mathbb{R}^n$$

Sufficient and necessary condition

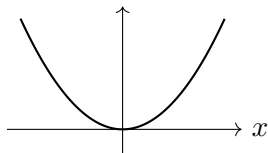
■ PSD HESSIAN

$$\nabla^2 f(x) \succeq 0, \quad \forall x \in \mathbb{R}^n$$

Sufficient and necessary condition

Scalar examples

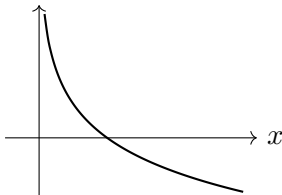
$$f(x) = x^2$$



$$f''(x) = 2 > 0$$

Strictly convex on $\mathbb{R}$

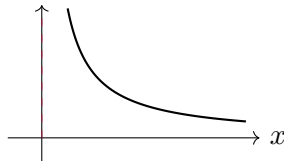
$$f(x) = -\log x$$



$$f''(x) = \frac{1}{x^2} > 0$$

Strictly convex on $x > 0$

$$f(x) = 1/x$$



$$f''(x) = \frac{2}{x^3} > 0$$

Strictly convex on $x > 0$

DOMAIN MATTERS: Convexity is always stated on a convex domain

Affine and quadratic functions

■ AFFINE FUNCTION

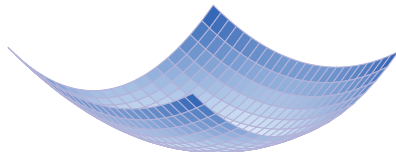
$$f(x) = a^\top x + b, \quad \nabla^2 f(x) = 0$$

- ★ affine functions are both convex and concave

■ QUADRATIC FUNCTION

$$f(x) = \frac{1}{2}x^\top Qx + b^\top x + c, \quad Q = Q^\top$$

- ★ $Q \succeq 0$ – sufficient and necessary condition



Bowl-shaped surface

Norms are convex functions

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

■ NORM

- ★ $f(x) \geq 0$ and $f(x) = 0$ iff $x = 0$ – non-negativity
- ★ $f(\alpha x) = |\alpha|f(x)$, $\forall \alpha \in \mathbb{R}$ – homogeneity
- ★ $f(x + y) \leq f(x) + f(y)$ – triangle inequality

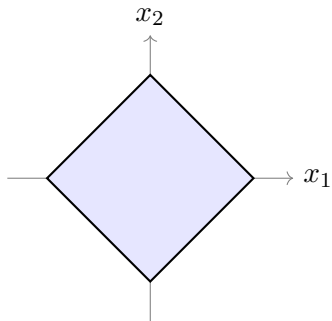
CONVEXITY PROOF: Homogeneity and Triangle inequality

■ ℓ_q -NORM

$$\|x\|_q = \left(\sum_{i=1}^n |x_i|^q \right)^{1/q}, \quad q \geq 1$$

- ★ $\|x\|_0 = \#\{i : x_i \neq 0\}$ – not a norm

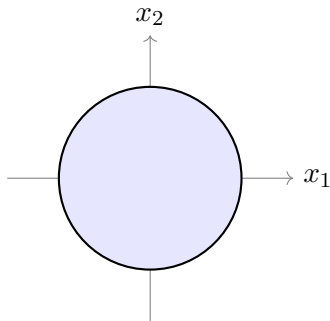
Geometry of norm balls



ℓ_1 -ball

$$\|x\|_1 \leq 1$$

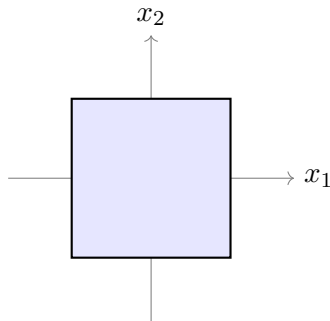
Sparsity



ℓ_2 -ball

$$\|x\|_2 \leq 1$$

Rotational symmetry



ℓ_∞ -ball

$$\|x\|_\infty \leq 1$$

Coordinate bounds

Closure rules: Build new convex functions

Convex functions: $f, f_i: \mathbb{R}^n \rightarrow \mathbb{R}$

■ NONNEGATIVE SCALING

$\alpha f(x)$ is convex, $\alpha \geq 0$

■ SUMS

$\sum_i w_i f_i(x)$ is convex, $w_i \geq 0$

■ AFFINE PRECOMPOSITION

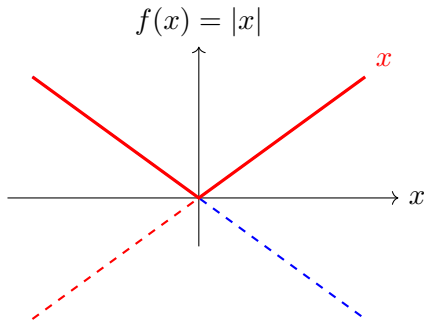
$g(x) = f(Ax + b)$ is convex

Pointwise supremum preserves convexity

Convex functions: $f_\alpha: \mathbb{R}^n \rightarrow \mathbb{R}, \alpha \in \mathcal{A}$

■ POINTWISE SUPREMUM

$$f(x) = \sup_{\alpha \in \mathcal{A}} f_\alpha(x) \text{ is convex}$$



Composition rules

Function: $g: \mathbb{R}^n \rightarrow \mathbb{R}$

Function: $h: \mathbb{R} \rightarrow \mathbb{R}$

■ COMPOSITE FUNCTION

$f(x) = h(g(x))$ is convex, when

g	h
convex	convex and nondecreasing
concave	convex and nonincreasing
affine	convex

- ★ $e^{g(x)}$ is convex, if g is convex
- ★ $-\log(g(x))$ is convex, if $g > 0$ is concave

Convex functions of matrices

■ AFFINE TRACE FUNCTION

$$f(X) = \text{Tr}(A^\top X) + b$$

■ LARGEST EIGENVALUE OF PSD MATRICES

$$\lambda_{\max}(X) = \sup_{\|u\|_2=1} u^\top X u, \quad X = X^\top \succeq 0$$

■ LARGEST SINGULAR VALUE

$$\sigma_1(X) = \|X\|_2 = \sup_{\|u\|=\|v\|=1} u^\top X v, \quad X \in \mathbb{R}^{m \times n}$$

■ KY FAN NORMS

$$\|X\|_{(k)} = \sum_{i=1}^k \sigma_i(X)$$

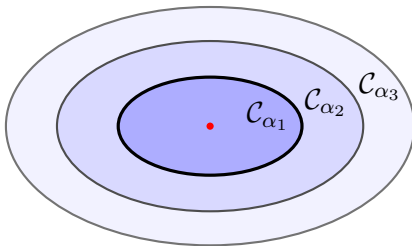
★ $k = \min\{m, n\}$ – nuclear norm

From convex functions to convex sets

Convex function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

■ SUBLEVEL SET

$\mathcal{C}_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}$ is convex

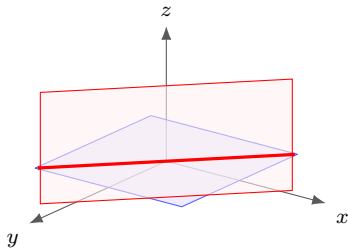


The converse is false

Examples of convex sets

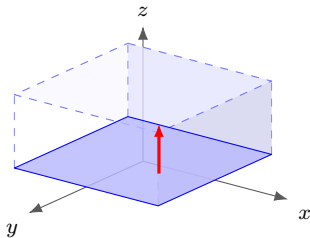
Affine set

$$\mathcal{A} = \{x \in \mathbb{R}^n \mid Ax = b\}$$



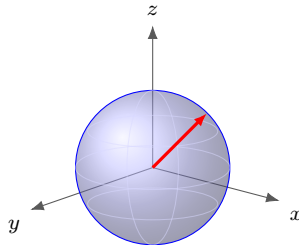
Halfspace

$$\mathcal{H} = \{x \in \mathbb{R}^n \mid a^\top x \leq b\}$$



Norm ball

$$\mathcal{B} = \{x \in \mathbb{R}^n \mid \|x - c\| \leq r\}$$



Intersection and polyhedra

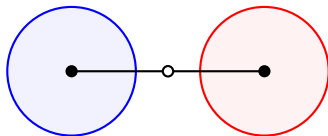
Convex sets: $\mathcal{C}_\alpha, \alpha \in \mathcal{A}$

■ INTERSECTION

$\bigcap_{\alpha \in \mathcal{A}} \mathcal{C}_\alpha$ is convex

★ $\{x \in \mathbb{R}^n \mid Ax \leq b\} = \bigcap_i \{x \in \mathbb{R}^n \mid a_i^\top x \leq b_i\}$ – polyhedron (finitely many halfspaces)

Union: generally non-convex



$\mathcal{C}_1 \cup \mathcal{C}_2$

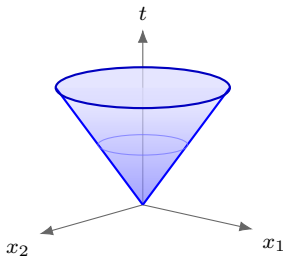
Convex cone

Convex set: $\mathcal{C} \subseteq \mathbb{R}^n$

■ CONVEX CONE $\mathcal{C}$

$$\alpha x \in \mathcal{C}, \quad \forall x \in \mathcal{C}, \alpha \geq 0$$

- ★ $\mathbb{R}_+^n$ – nonnegative orthant
- ★ $\{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid \|x\| \leq t\}$ – quadratic cone



- ★ $\{X \in \mathbb{R}^{n \times n} \mid X = X^\top \succeq 0\}$ – PSD cone

Affine images and preimages

Convex set: $\mathcal{C} \subseteq \mathbb{R}^n$

Affine transformation: $T(x) = Ax + b$

■ AFFINE IMAGE

$T(\mathcal{C}) = \{Ax + b \mid x \in \mathcal{C}\}$ is convex

■ AFFINE PREIMAGE

$T^{-1}(\mathcal{C}) = \{x \in \mathbb{R}^n \mid Ax + b \in \mathcal{C}\}$ is convex

★ $\{x \in \mathbb{R}^n \mid A_0 + \sum_{i=1}^n x_i A_i \succeq 0\}$, $A_i = A_i^\top$ – feasible set of linear matrix inequalities

Questions?

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